

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2024 Course: B. Tech. Branch: All Semester :I Subject Code & Name: Engineering Physics (BTBS102P) Max Marks: 60 Date: 02/07/2024 Duration: 3 Hr.			
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)		Marks
Q. 1	Solve Any Two of the following.			12
A)	What is free vibration? Derive an expression for differential equations for free oscillation.	(CO1) (Remember & Understand)		6
B)	Define resonance with one example. Explain i) Amplitude of Resonance ii) Sharpness of resonance.	(CO1) (Understand)		6
C)	State any four applications of ultrasonic waves. A quartz crystal having 5 mm thickness is vibrating at resonance. Calculate the fundamental frequency of vibrations for which ultrasonic waves are generated. Given for quartz, Young's Modulus is 7.9×10^{10} N/m ² , Density is 2650 Kg/m ³	(CO1) (Remember & Understand)		6
Q.2	Solve Any Two of the following.			12
A)	Explain construction and working of Ruby laser.	(CO2) (Understand)		6
B)	State & explain Brewster's law of polarization. A glass plate is to be used as a polarizer. Find the angle of polarization for it. ($\mu = 1.54$)	(CO2) (Remember & Understand)		6
C)	Explain types of optical fibers on the basis of refractive index and mode of propagation.	(CO2) Understand)		6

Q. 3	Solve Any Two of the following.		12
A)	Explain the construction and working of Bainbridge Mass Spectrograph with neat and labeled diagram.	(CO3) (Understand)	6
B)	State Heisenberg's uncertainty principle. Using this prove that electron cannot exist in the nucleus.	(CO3) (Remember & Understand)	6
C)	Derive Schrodinger's time dependent wave equation.	(CO3) (Understand)	6
Q.4	Solve the following questions.		12
A)	Derive the relation between lattice constant (a) and density (ρ) of material. A substance with face centered cubic lattice has density 6250 kg/m^3 and molecular weight 60.2. Calculate the lattice constant a. Given Avogadro's number $=N= 6.023 \times 10^{26}$ per kg moles.	(CO4) (Understand)	6
B)	Distinguish between Continuous and Characteristics X-ray spectra.	(CO4) (Understand)	6
Q. 5	Solve Any Two of the following.		12
A)	What is superconductivity? Explain Meissner effect in superconductors.	(Understand)	6
B)	Define Retentivity and Coercivity and explain B-H Curve.	(Remember & Understand)	6
C)	Explain Conductor, Semiconductor, and Insulator on the basis of energy band diagram.	(Remember & understand)	6
	*** End ***		